Design 1 Demo Project 2141 Icon Way, Vacaville, CA 95688, USA

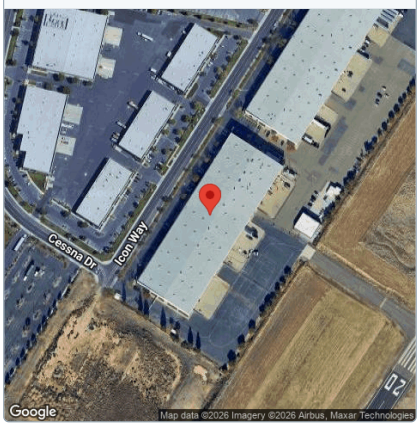
Project Details

Address	2141 Icon Way, Vacaville, CA 95688, USA
Customer Name	Demo Customer
Owner	Hesham Abas
Last Modified	Hesham Abas a few seconds ago
Location	(38.38471039662245, -121.95944951654359) (GMT -8)
Profile	Demo Project Profile

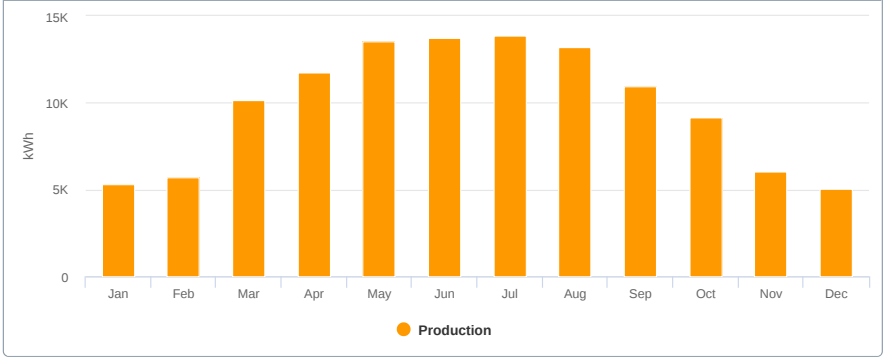
System Metrics

Design	Design 1
Module DC Nameplate	68.40 kW
Inverter AC Nameplate	60.00 kW Load Ratio: 1.14
Annual Production	118.5 MWh
Performance Ratio	83.7%
kWh/kWp	1,732.5
Weather Dataset	TMY, 10km grid (38.35,-121.95), NREL (prospector)
Simulator Version	6a07bb85fa-d41d5e5ba5-217e7279bb-252e91d342

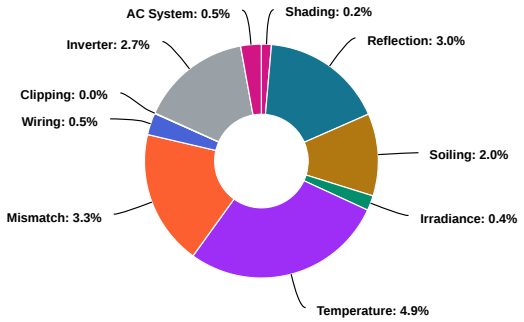
Project Location



Energy Consumption



System Loss



Annual Production

	Description	Output	% Delta
Irradiance (kWh/m²)	Annual Global Horizontal Irradiance	1,875.6	-
	POA Irradiance	2,070.4	10.4%
	Shaded Irradiance	2,065.3	-0.2%
	Irradiance After Reflection	2,003.6	-3.0%
	Irradiance After Soiling	1,963.5	-2.0%
	Total Collector Irradiance	**1,963.5**	**0.0%**
Energy (kWh)	Nameplate	134,331.5	-
	Output at Irradiance Levels	133,844.6	-0.4%
	Output at Cell Temperature Derate	127,253.9	-4.9%
	Output after Electrical Mismatch	123,077.8	-3.3%
	Optimal DC Output	122,423.7	-0.5%
	Constrained DC Output	122,405.1	-0.0%
	Inverter Output	119,099.7	-2.7%
	AC System Output	118,504.2	-0.5%
	Energy to Grid	**118,504.2**	**0.0%**

Temperature Metrics	
Avg. Operating Ambient Temp	19.3°C
Avg. Operating Cell Temp	29.9°C

Simulation Metrics	
Operating Hours	4,681

Condition Set

Description	Condition Set 1
Weather Dataset	TMY, 10km grid (38.35,-121.95), NREL(prospector) (download)
Solar Angle Location	Meteo Lat/Lng
Transposition Model	Perez Model
Temperature Model	Sandia Model

Temperature Model Parameters

Rack Type	a	b	Temperature Delta
Fixed Tilt	-3.56	-0.08	3.0°C
Flush Mount	-2.81	-0.05	0.0°C
East-West	-3.56	-0.08	3.0°C
Carport	-3.56	-0.08	3.0°C

Soiling (%)

J	F	M	A	M	J	J	A	S	O	N	D
2	2	2	2	2	2	2	2	2	2	2	2

Albedo

J	F	M	A	M	J	J	A	S	O	N	D
0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3

Rear Mismatch Loss	10%	**Rear Shading Factor**	5%
Module Transparency	2%		
Irradiation Variance	5.0%		
Cell Temperature Spread	4.0°C		
Module Binning Range	-2.5% to 2.5%		
AC System Derate	0.50%		

Component Characterizations

Type	Component	Characterization	Bifacial
Module	JKM380M-6RL3-V (1000V) (Jinko)	Spec Sheet Characterization, PAN	False
Inverter	SH6T (380V) (Sungrow)	Spec Sheet	N/A

Design BOM

Component	Type	Specs
1 input Combiners	Combiners	20 (-)
SH6T (380V) (Sungrow)	Inverters	10 (60.00 kW)
Jinko, JKM380M-6RL3-V (1000V), (380W)	Modules	180 (68.40 kW)

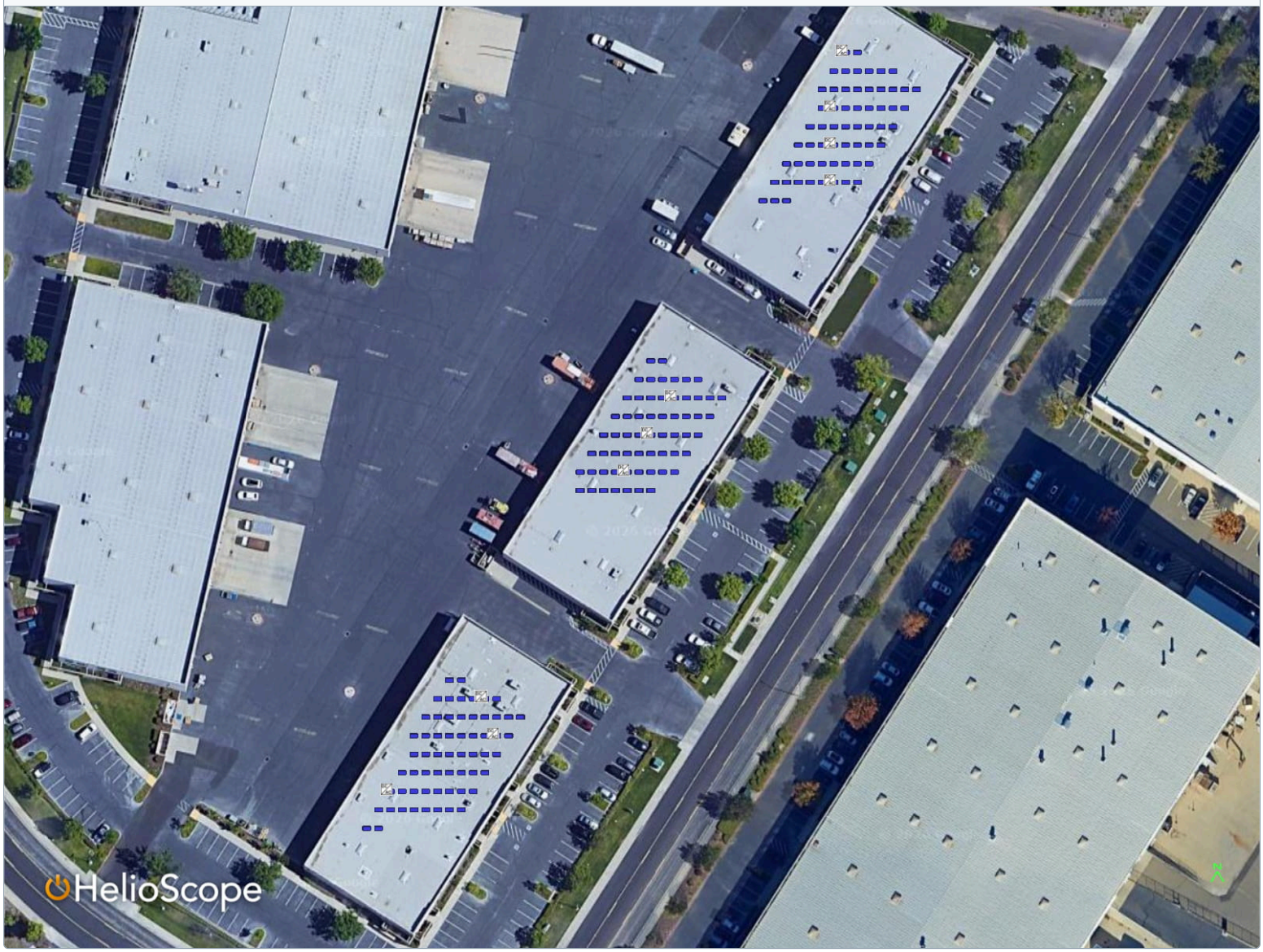
Monthly Shading

Month	GHI (kWh/m^2)	POA (kWh/m^2)	Shaded (kWh/m^2)	Nameplate (kWh)	Grid (kWh)
January	68.1	88.8	88.6	5,706.1	5,284.1
February	79.4	96.2	96.0	6,195.6	5,688.8
March	151.2	173.2	172.8	11,244.3	10,178.5
April	186.8	201.0	200.5	13,070.7	11,717.1
May	231.6	236.9	236.3	15,415.1	13,563.2
June	244.9	244.8	244.2	15,932.5	13,747.2
July	246.7	249.8	249.1	16,259.4	13,899.8
August	222.5	235.9	235.3	15,358.5	13,195.4
September	172.6	195.8	195.3	12,719.4	10,945.3
October	129.6	159.3	158.9	10,313.0	9,172.4
November	78.5	103.5	103.3	6,657.1	6,050.4
December	63.7	85.2	85.0	5,459.9	5,061.9

Design Wiring Zone

Description	Combiner Poles	String Size	Stringing Strategy
Wiring Zone	12	5 - 21	Along Racking

Design Render



Field Segments

Description	Racking	Orientation	Tilt	Azimuth	Intrarow Spacing	Frame Size	Frames	Modules	Power
Field Segment 1 (copy)	Fixed Tilt	Landscape (Horizontal)	Module: 15°	Module: 180°	11.0 ft	1x1	60	60	22.80 kW
Field Segment 1 (copy 1)	Fixed Tilt	Landscape (Horizontal)	Module: 15°	Module: 180°	11.0 ft	1x1	60	60	22.80 kW
Field Segment 1 (copy 2)	Fixed Tilt	Landscape (Horizontal)	Module: 15°	Module: 180°	11.0 ft	1x1	60	60	22.80 kW